

3. (a) A student is investigating the current-voltage characteristic of a filament lamp. In the space below, **draw a circuit diagram** of the circuit she would use. [3]

(b) The student collects the following results.

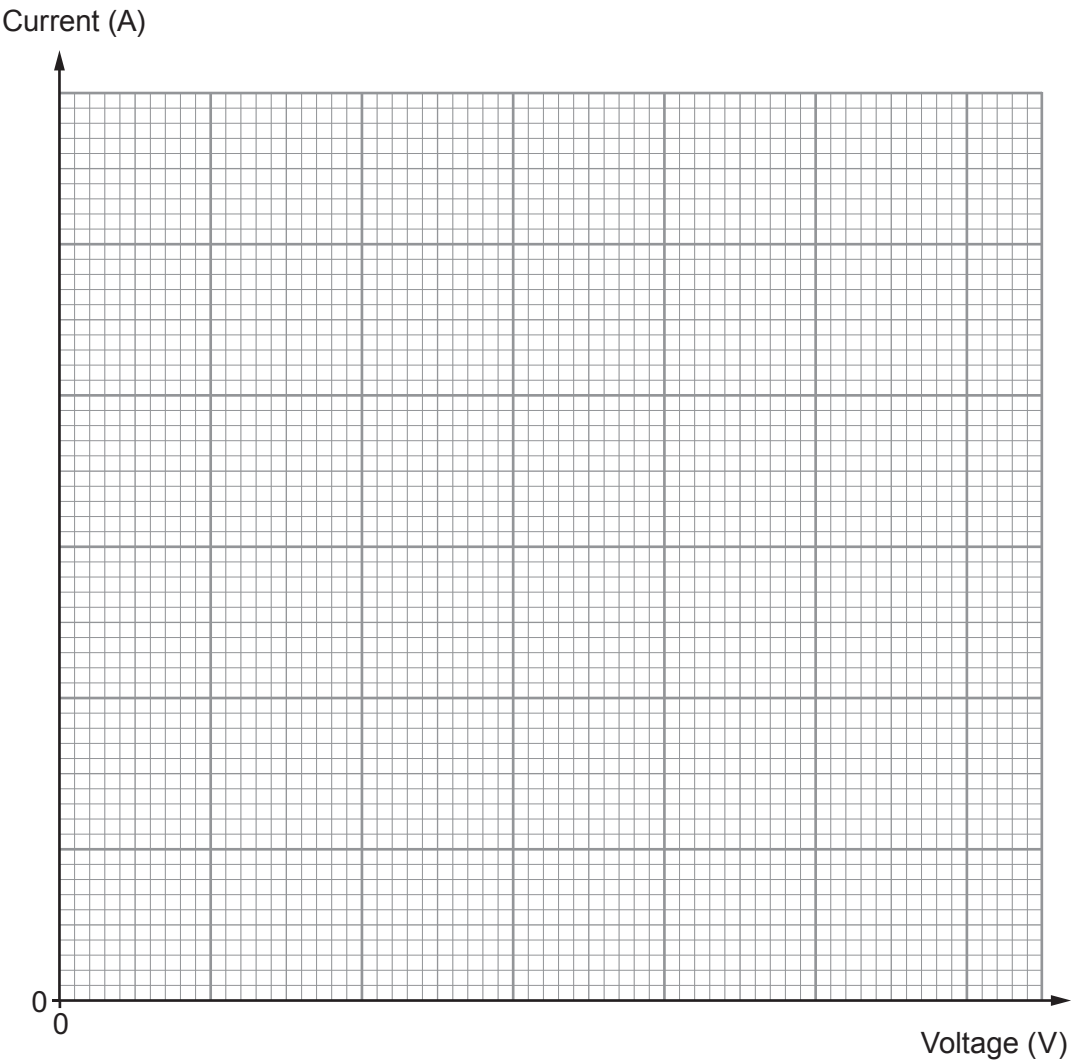
Voltage (V)	Current (A)
0.0	0.00
2.0	0.40
4.0	0.76
6.0	0.94
10.0	1.08
12.0	1.10



Examiner
only

(i) Plot the data on the grid below and draw a suitable line.

[3]



(ii) She concludes that the resistance of the lamp decreases as voltage increases. Explain using calculations, whether you agree or disagree with this conclusion. [3]

Space for calculations

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